

Vapor–Liquid–Solid-Mediated Layer-by-Layer Growth of Stepped-Wedge-Shaped WS₂ Microribbons Using the Chemical Vapor Deposition Technique

Umakanta Patra, Faiha Mujeeb, and Subhabrata Dhar*



Cite This: *Cryst. Growth Des.* 2024, 24, 1626–1631



Read Online

ACCESS |



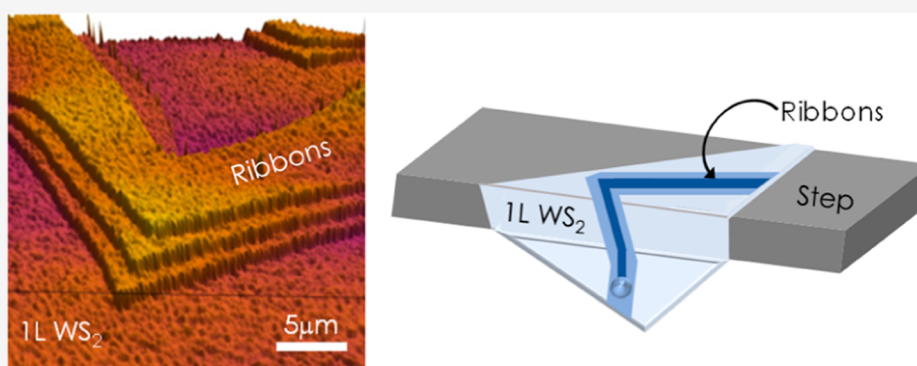
Metrics & More



Article Recommendations



Supporting Information



ABSTRACT: The bottom-up approach of fabricating semiconductor microstructures is more cost-effective and time-efficient than the conventional top-down method of lithography. However, gaining control over the size, shape, position, and directionality of the structure is a big challenge. Controlled growth of microstructures of two-dimensional (2D) transition-metal dichalcogenides, which are emerging as promising materials for transparent and flexible electronics and valleytronics, has gained significant research interest recently. Here, we report a unique vapor–liquid–solid (VLS)-mediated growth, where the VLS process repeats itself multiple times along the same path that results in the formation of several tens of micrometer-long stepped-wedge-shaped WS₂ microribbons of different widths and thicknesses. These ribbons are found to be vertically stacked WS₂ monolayer stripes whose width decreases from the bottom to the top. It has been found that back-and-forth motion of the Na–W–O droplets that result from the mixture of precursor WO₃ and promotor NaCl over a WS₂ monolayer island along certain crystallographic directions multiple times results in the layer-by-layer growth of these ribbons. This study thus allowed us to determine a novel bottom-up route for constructing microstructures using 2D materials with shape, size, position, and orientation specificity.

INTRODUCTION

Transition-metal dichalcogenides (TMDCs), which are two-dimensional (2D)-layered semiconductors, have great potential for electronic and optoelectronic device applications.^{1–4} The class of materials exhibits several fascinating properties like large exciton binding energy;⁵ existence of many body bound states, e.g., trions⁶ and biexcitons;⁷ and thickness dependence of the electronic properties.⁸ Strong spin-valley coupling in monolayers also leads to the valley polarization effect,^{9,10} which can be exploited to inject spin-polarized carriers and excitons for future spintronics and valleytronics. Due to their layered nature, lower dielectric constant, higher electron effective mass, and larger band gap than that of silicon, TMDCs are superior in suppressing the short channel effects that facilitate greater integration of devices in a chip. These materials also offer high elastic constants and large breaking strengths, which make them suitable for flexible electronics. Among the TMDCs, WS₂ shows the highest carrier mobility at

room temperature (1000 cm²/V-s). It also has a large spin–orbital coupling energy of ~420 meV.¹¹ These properties make WS₂ a potential candidate for device applications. Field-effect transistors (FETs) have been demonstrated on both mono- and multilayers of various TMDC materials.^{12–14} It should be noted that, as far as the performance of the FET is concerned, multilayers have certain advantages over monolayers.¹⁵

Due to the larger surface-to-volume ratio in monolayers, their electrical conductivity is poorer than that of the multilayers.¹⁴ Moreover, the direct nature of the bandgap in monolayers can result in a higher level of thermal noise.¹⁶

Received: October 23, 2023

Revised: January 26, 2024

Accepted: January 26, 2024

Published: February 5, 2024



Fabrication of devices is commonly carried out following top-down routes, which involve several lithographic processing steps. The bottom-up approach, on the other hand, is more economical and time-efficient. However, in this methodology, control over the shape, size, location, and orientation of the grown structure is the main challenge. Vapor–liquid–solid (VLS) mode of growth might be the key to achieve such a control. In fact, there are reports of the synthesis of ribbons of various TMDCs using the VLS mechanism.^{17,18} In certain cases, these ribbons are found to grow along certain directions on the monolayer flakes.¹⁹

Here, we have shown that layer-by-layer growth of stepped-wedge-shaped WS₂ microribbons of varied widths and thicknesses that run along certain specific directions is possible on SiO₂-coated Si substrates via a VLS methodology using the chemical vapor deposition (CVD) technique. These ribbons, which are formed on top of triangular WS₂ monolayer flakes, run parallel to each other from one edge to another edge of the flake. When examined closely, these structures are found to be vertically stacked of WS₂ monolayer stripes, whose width decreases from the bottom to the top, giving the ribbon the shape of a stepped wedge. This study suggests that Na–W–O droplets, where Na ions are sourced from NaCl used as the growth promoter, leave behind the trails of the WS₂ monolayer ribbon, while the droplet physically moves over the monolayer flake. To-and-fro motion of the droplets multiple times results in the growth of these stepped-wedge-shaped ribbon structures. This study thus reveals a new type of VLS mechanism, where the process repeats itself multiple times along the same path, leading to the step-by-step construction of the microstructure.

EXPERIMENTAL SECTION

Samples were grown in a CVD chamber, which is depicted schematically in Supporting Information Figure S4. A ceramic boat containing 30 mg of high-purity WO₃ powder (99.995%, Aldrich) mixed with 2 mg of NaCl powder (99.5%, Aldrich) was kept in the hottest zone of the quartz reactor tube, while another ceramic boat with 350 mg of sulfur (S) powder (99.95%, Aldrich) was placed in a cooler zone in the reactor. The separation between the two boats was maintained at 18.5 cm. The SiO₂/Si substrate was cleaned successively in trichloroethylene (TCE), acetone, and methanol for 5 min each in an ultrasonic bath before being placed on the floor of the WO₃-containing boat, as shown in Figure S4.1a. The quartz reactor was purged with argon (Ar) gas for 10 min at a flow rate of 300 sccm at room temperature, followed by Ar-purging for 20 min at a flow rate of 200 sccm at a furnace temperature of 150 °C. Then, the flow rate of Ar was reduced to 8 sccm, and the furnace was heated to 1000 °C at a rate of 10 °C/min. This temperature was maintained for 10 min. At this condition, the S boat attained about 150 °C. The furnace was then allowed to cool with the same flow rate of Ar until it reached 650 °C. Beyond this point, the flow rate was increased to 500 sccm and maintained until the temperature attained 300 °C. The flow rate was then changed to 50 sccm, and finally it was stopped once the temperature was below 100 °C. The temperature evolutions at the WO₃ and S boat positions are shown in Supporting Information Figure S4.1b.

CHARACTERIZATION

Optical images and atomic force microscopy (AFM) of the sample were performed at room temperature. Photoluminescence (PL) and Raman spectroscopy were carried out using a micro-PL/Raman setup from Renishaw, Invia Reflex, UK. PL and Raman maps were recorded with 50× magnification and a 1 μm scanning step. Raman and PL spectroscopy were carried

out using a 532 nm diode laser excitation source with 500 μW power and a 500 cm focal-length monochromator equipped with a 2400 gr/mm grating and CCD detector.

RESULTS AND DISCUSSION

Growth is carried out in a CVD chamber with WO₃ and sulfur powders as the precursors for W and S, NaCl as the growth promoter, and SiO₂-coated Si as the substrate. Figure 1a,b

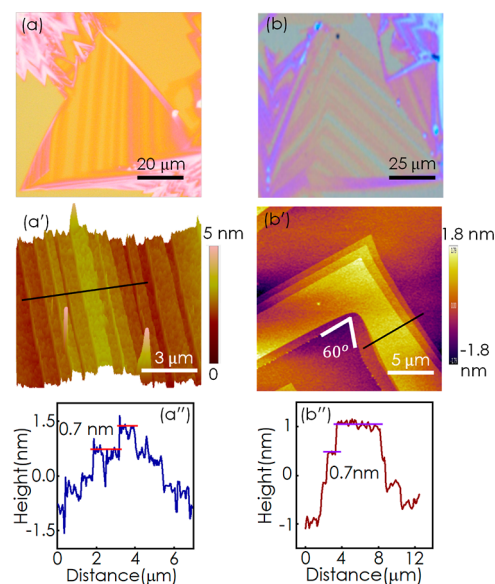


Figure 1. Optical images of triangular flakes decorated with (a) parallel straight ribbons and (b) parallel V-shaped ribbons. AFM image taken on (a') straight and (b') V-shaped ribbons. (a'',b'') represent the AFM line-scan profiles recorded along the black lines shown in images (a',b'), respectively.

represents the optical images of two types of triangular flakes recorded from different parts of the samples. These flakes are decorated with either straight or V-shaped parallel strips. It is particularly interesting to note that all V-shaped stripes in Figure 1b have a bending angle of 60°. In both images, other structures with much brighter contrasts are also visible. These features are present mostly at the edges or outside the boundary of the triangular flake, which are likely to be the monolayers of WS₂, while the regions of higher contrasts can be attributed to thicker WS₂ deposits. AFM images taken on the stripes are presented in Figure 1a',b', respectively, while the line scans recorded along the black lines shown in the respective images are plotted in Figure 1a'',b''. It is quite evident that both types of stripes (straight and V-shaped) have stepped-wedge structures with a step height of ~0.7 nm. This value matches very well with the thickness of the WS₂ monolayer.²⁰ It is noteworthy that the stripes of different contrasts and widths can be seen in Figure 1a,b. AFM investigation on these structures reveals that they all have a stepped-wedge-shaped ribbon structure with a uniform step height of ~0.7 nm. However, the number of steps is different in different ribbons. One more point has also been noticed that the ribbons are always extended from one edge to the other of the underlying WS₂ flakes.

Figure 2a shows an AFM image of a spiral structure recorded from some other part of the sample. The spiral is formed as the ribbon, in this case, has taken several turns. It has been found that the angle of bending is always an integer multiple of 60°. It

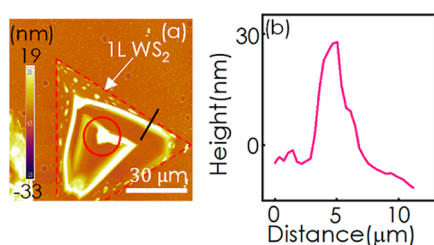


Figure 2. (a) AFM image of spirally grown flat-edge microribbons. (b) AFM line-scan profile taken along the black line shown in image (a).

is also evident from the figure that the structure is formed on a triangular island. Figure 2b shows the AFM line-scan profile taken over the black line shown in Figure 2a. Interestingly, the spiral ribbon does not seem to have any stepped edge (Figure 2c',c'') structure. Moreover, the thickness of the ribbons is also much greater than that of the straight or V-shaped parallel ribbons.

Figure 3 shows the results of the Raman study carried out on the triangular flake decorated with straight parallel ribbons.

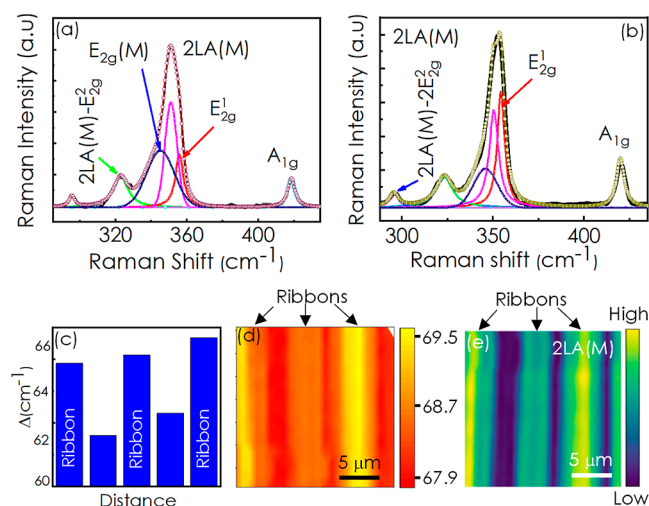


Figure 3. Representative Raman spectra recorded from a point that is located (a) between the ribbons and (b) on the ribbon. Deconvolution of the profiles with six Lorentzian functions is also presented. (c) Average separation between E'_{2g} and A_{1g} peaks (Δ) on several ribbons and the regions between the ribbons. (d) 2LA(M) to the A_{1g} peak-separation map. (e) Map of the 2LA(M) peak intensity.

Figure 3a,b presents the Raman profiles recorded on the points located between the ribbons and on a ribbon, respectively. In both cases, the Raman spectrum can be deconvoluted with six Lorentzian functions, as shown in the figures. The six peaks can be identified as an out-of-plane vibration peak (A_{1g}), an in-plane vibration peak (E'_{2g}), the second order of longitudinal acoustical phonon mode at the M-point [2LA(M)], the E_{2g} mode at the M-point [$E_{2g}(M)$], as well as the coupled modes 2LA(M) – $E_{2g}^2(\Gamma)$ and 2LA(M) – $E_{2g}^2(\Gamma)$ of WS_2 .²¹

In both profiles, the 2LA(M) mode is the most dominant peak. Note that the separation between the in-plane (E'_{2g}) and out-of-plane (A_{1g}) vibration peaks depends strongly on the layer number.²² Figure 3c plots the average separation between the E'_{2g} and A_{1g} peaks on several ribbons and the regions between them. Evidently, peak separation for the regions between the ribbons ranges between 62 and 63 cm^{-1} , which is

expected for monolayer WS_2 ,²³ while the separation varies between 65 and 67 cm^{-1} for the areas under the ribbons. A 2LA(M) to A_{1g} peak-separation map covering several straight ribbons is presented in Figure 3d. The clear contrast between the regions with and without ribbons is quite evident. All of these clearly suggest that the ribbons are multiple layers of WS_2 decorated on a monolayer WS_2 flake, which is further corroborated by the findings of Figure 1. Figure 3e shows the Raman intensity map for the 2LA(M) peak of the same region. Higher intensity is resulting from the regions with ribbons as multiple layers are involved in Raman scattering.²⁴ The intensity contrast in the picture further evidences the stepped-wedge nature of these ribbons, as observed from the AFM study. It should be noted that similar Raman investigations are carried out on the triangular flake decorated with parallel V-shaped ribbons and on the thicker flat-edge (nonstepped-wedge) ribbons (like the spiral one shown in Figure 2a). The results for the V-shaped ribbons are found to be very similar to those of the parallel straight ribbons. Thicker flat-edge ribbons are also found to be multilayer WS_2 (see Supporting Information S1). Figure 4 represents the results of

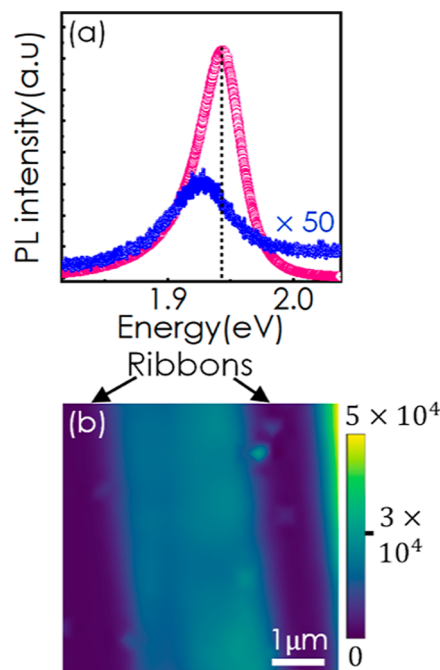


Figure 4. (a) Room-temperature PL spectra recorded on the triangular flake decorated with straight, parallel ribbons. Red and blue profiles represent the spectra recorded from a point between the ribbons and on the ribbons, respectively. (b) PL intensity map recorded over a region partly covering two ribbons.

the room-temperature PL study for the triangular flake decorated with straight parallel ribbons. PL spectra obtained from two points, one located between the ribbons and another directly on a ribbon, are compared in Figure 4a. Evidently, the PL intensity is much higher for the region between the ribbons than that obtained on the ribbon. Further, the PL peak for the region between the ribbons appears at 1.94 eV, which matches very well with the peak position of neutral A-excitons in monolayer WS_2 , as reported in the literature.²⁰ However, the peak appears at a red-shifted position of 1.92 eV when recorded on the ribbon. This implies that the ribbons are made of multiple layers of WS_2 .²⁵ It should be noted that triangular

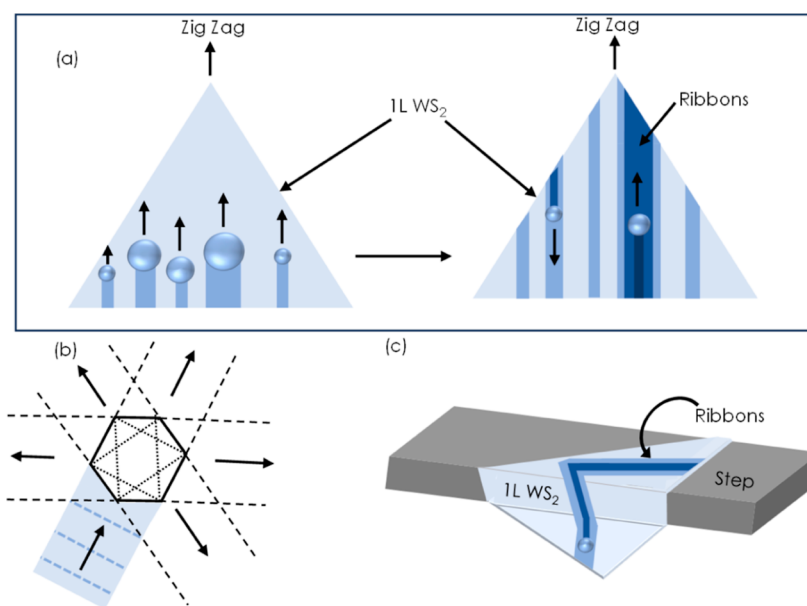


Figure 5. Schematic depiction of (a) VLS-mediated layer-by-layer growth of stepped-wedge-shaped WS_2 microribbons on a monolayer flake, (b) six equivalent zigzag directions along which ribbon growth can proceed, and a (c) probable mechanism for the growth of 60-degree bended (V-shaped) stepped-wedge-shaped WS_2 microribbons.

flakes, which are not decorated with any ribbon, are also found on the sample. AFM and Raman studies confirm that these flakes are monolayer WS_2 (see Supporting Information S2). Room-temperature PL spectra recorded on these flakes also show a strong peak at 1.94 eV. All of these results are thus consistent with the fact that the triangular flakes, on which the ribbons are decorated, are monolayer WS_2 . Figure 4b presents the PL intensity map for a region partially covering two ribbons. The sharp contrast between the region with and without ribbons is quite evident. The drastic reduction of PL intensity on the ribbons is consistent with their multilayer nature.²⁶ It should be noted that a micro-PL study on the V-shaped stripes also shows similar results. Micro-PL studies are also carried out on the spiral ribbons. The results confirm that they are made of multilayers of WS_2 .

Note that NaCl, which has been used as a growth promoter in this case, can form the intermediate compound Na_2WO_4 after reacting with WO_3 .²⁷ It is highly likely that the intermediate compound with a low melting point leads to the formation of Na–W–O droplets on the substrate surface and helps in the growth of WS_2 ribbons via the VLS pathway.¹⁹ The stepped-wedge shape of the ribbons in Figure 1a,b suggests that these structures must have been grown in a layer-by-layer process, which means that the VLS growth has been repeated multiple times as a result of back-and-forth movement of the droplet along the same path. This is shown schematically in Figure 5a. It is plausible that the monolayer ribbon formation is first initiated by flat droplets developing at the edge of the monolayer flakes, which are likely to be grown on the substrate before the casting of the droplets takes place. As time progresses, a monolayer stripe grows out of the droplet, and the latter moves along a certain direction over the flake, leaving behind a trail of a monolayer ribbon of width comparable to the diameter of the droplet. It is possible that, like monolayer MoS_2 , the zigzag direction of WS_2 may also have a higher growth rate than that of the armchair direction.^{19,28} This can lead to the growth of the ribbons only along certain crystalline directions of WS_2 , as shown

schematically in Figure 5b. When the droplet reaches another edge of the flake, its forward motion is obstructed, and it is compelled to trace back its path in the opposite direction. This process continues multiple times until the droplet is fully used up (see Supporting Information Figure S3.1, where an abrupt ending of the top layer of a ribbon is captured in the AFM image). It should be noted that in some of the optical and AFM images, we have indeed found the remains of the droplets at the end points of the top layers in stepped microribbons, as shown in Supporting Information Figure S3.2. In order to check the composition, we have performed energy dispersive X-ray spectroscopy (EDS) on these droplets. Supplementary Figure S3.3 shows the EDS spectrum and a table containing the weight percentage of the elements present in one such droplet. The presence of tungsten (W), oxygen (O), sodium (Na), and sulfur (S) has been found. All these observations strongly suggest that the growth of the stepped microribbons takes place through a layer-by-layer VLS pathway. It is plausible that every time the droplet hits the edge, a part of it is lost, which means that its size decreases. This might be the reason why the width of the layers in the stack decreases from the base to the top, as observed in the AFM investigation (see Figure 1). Since the WS_2 lattice has hexagonal symmetry, the zigzag (or armchair) direction has 6-fold symmetry. If the growth of the ribbon is obstructed due to a certain perturbation even before the droplet reaches the edge of the underlying monolayer flake, the growth front can switch its direction to one of the five other equivalent directions that are 60° separated from each other, as schematically depicted in Figure 5b. This can explain the 60° bending observed in V-shaped stripes. It seems that the triangular flake that is decorated with V-shaped stripes lies on a naturally present step on the substrate, as marked in Figure 1b. This step might be acting as a perturbation that instigates the growth front to change its direction by 120°, as schematically depicted in Figure 5c. Interestingly, the shape of the thicker nonstepped ribbons suggests that, unlike the stepped ribbons, these ribbons are grown in a one-step VLS route. We believe that the growth

of these ribbons is initiated by the fatter (more spherical) Na–W–O droplets, which are formed in the interior of the underlying WS₂ island. It is plausible that the wetting property of the droplet material with WS₂ favors the formation of fatter droplets than those formed at the edge of the island. The VLS process leads to the growth of multilayer ribbons running along the zigzag (or armchair) directions. These ribbons can occasionally make a 60° bend, as can be seen in the case of the spiral ribbon presented in Figure 2. It could be possible that, in the case of the spiral structure shown in Figure 2, the droplet breaks into two pieces at the starting nucleation point. The bigger and smaller droplets run ribbons of different thicknesses and widths along the opposite directions. Figure 6

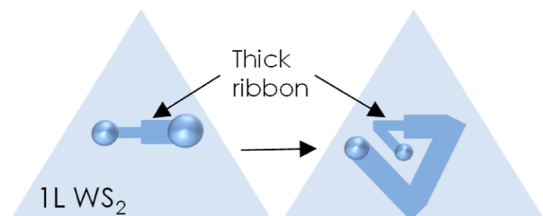


Figure 6. Schematic picture of the VLS growth of nonstepped thicker ribbons forming a spiral.

depicts the mechanism schematically. In Figure 2a, a location is marked with a red circle, which we believe is the nucleation starting and droplet breaking point, as the dimensions of the ribbon at that point change abruptly. Note that the nucleation starting point is evidently inside the underlying triangular island.

CONCLUSIONS

WS₂-stepped ribbons of varied widths, heights, and shapes are possible to grow on SiO₂-coated Si substrates via the VLS methodology. Ribbons made of a stack of WS₂ monolayers, whose width decreases from the bottom to the top, giving them the shape of a stepped wedge, are found to be formed on top of the WS₂ monolayer flakes. This study suggests that to-and-fro motion of Na–W–O droplets, where Na ions are sourced from NaCl used as the growth promoter, over a monolayer WS₂ flake along certain crystallographic directions multiple times results in the layer-by-layer growth of these stepped-wedge-shaped ribbon structures. The droplets that drive this type of growth are initially formed at the edge of the WS₂ monolayer flakes. VLS growth of thicker, nonstepped ribbons is also observed, which is attributed to fatter droplets formed inside the monolayer WS₂ islands. Occasional 60° bending has been found in both types of ribbons. This has been attributed to certain perturbations that lead to the switching of the growth front between six equivalent directions.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.cgd.3c01253>.

Detailed characterization of thick flat-edge microribbons, isolated monolayer triangle without microribbons, ending of the topmost ribbon in the stepped microribbons, and schematic picture of the CVD growth setup (PDF)

AUTHOR INFORMATION

Corresponding Author

Subhabrata Dhar – Department of Physics, Indian Institute of Technology, Bombay, Mumbai 400076, India;
Email: dhara@phy.iitb.ac.in

Authors

Umakanta Patra – Department of Physics, Indian Institute of Technology, Bombay, Mumbai 400076, India; orcid.org/0009-0000-9966-9721

Faiha Mujeeb – Department of Physics, Indian Institute of Technology, Bombay, Mumbai 400076, India

Complete contact information is available at:
<https://pubs.acs.org/doi/10.1021/acs.cgd.3c01253>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We acknowledge the financial support from the Science and Engineering Research Board (SERB) under grant no. CRG/2022/001852, Government of India. We thank Jai Israni for his help in managing PL and Raman data. The authors are also thankful to the Industrial Research and Consultancy Centre (IRCC), the Sophisticated Analytical Instrument Facility (SAIF), and the Centre for Excellence in Nanoelectronics (CEN), IIT Bombay, for providing certain characterization facilities.

REFERENCES

- (1) Aggarwal, P.; Kaushik, S.; Bisht, P.; Sharma, M.; Singh, A.; Mehta, B. R.; Singh, R. Centimeter-Scale Synthesis of Monolayer WS₂ Using Single-Zone Atmospheric-Pressure Chemical Vapor Deposition: A Detailed Study of Parametric Dependence, Growth Mechanism, and Photodetector Properties. *Cryst. Growth Des.* **2022**, *22* (5), 3206–3217.
- (2) Mohapatra, P. K.; Deb, S.; Singh, B. P.; Vasa, P.; Dhar, S. Strictly Monolayer Large Continuous MoS₂ Films on Diverse Substrates and Their Luminescence Properties. *Appl. Phys. Lett.* **2016**, *108* (4), 042101.
- (3) Deb, S.; Chakrabarti, P.; Mohapatra, P. K.; Barick, B. K.; Dhar, S. Tailoring of Defect Luminescence in CVD Grown Monolayer MoS₂ Film. *Appl. Surf. Sci.* **2018**, *445*, 542–547.
- (4) Mohapatra, P. K.; Ranganathan, K.; Ismach, A. Selective Area Growth and Transfer of High Optical Quality MoS₂ Layers. *Adv. Mater. Interfaces* **2020**, *7* (24), 2001549.
- (5) Hanbicki, A. T.; Currie, M.; Kioseoglou, G.; Friedman, A. L.; Jonker, B. T. Measurement of High Exciton Binding Energy in the Monolayer Transition-Metal Dichalcogenides WS₂ and WSe₂. *Solid State Commun.* **2015**, *203*, 16–20.
- (6) Efimkin, D. K.; MacDonald, A. H. Many-Body Theory of Trion Absorption Features in Two-Dimensional Semiconductors. *Phys. Rev. B* **2017**, *95* (3), 035417.
- (7) Torche, A.; Bester, G. Biexcitons Fine Structure and Non-Equilibrium Effects in Transition Metal Dichalcogenides Monolayers from First Principles. *Commun. Phys.* **2021**, *4* (1), 67.
- (8) Son, Y.; Wang, Q. H.; Paulson, J. A.; Shih, C.-J.; Rajan, A. G.; Tvrdy, K.; Kim, S.; Alfeeli, B.; Braatz, R. D.; Strano, M. S. Layer Number Dependence of MoS₂ Photoconductivity Using Photocurrent Spectral Atomic Force Microscopic Imaging. *ACS Nano* **2015**, *9* (3), 2843–2855.
- (9) Chakrabarti, P.; Mujeeb, F.; Dhar, S. Enhancement of Valley Polarization in CVD Grown Monolayer MoS₂ Films. *Appl. Phys. Lett.* **2022**, *121* (7), 072103.
- (10) Mujeeb, F.; Chakrabarti, P.; Mahamiya, V.; Shukla, A.; Dhar, S. Influence of Defects on the Valley Polarization Properties of

Monolayer MoS₂ Grown by Chemical Vapor Deposition. *Phys. Rev. B* **2023**, *107* (11), 115429.

(11) Lan, F.; Yang, R.; Hao, S.; Zhou, B.; Sun, K.; Cheng, H.; Zhang, S.; Li, L.; Jin, L. Controllable Synthesis of Millimeter-Size Single Crystal WS₂. *Appl. Surf. Sci.* **2020**, *504* (May 2019), 144378.

(12) Chen, J.; Shao, K.; Yang, W.; Tang, W.; Zhou, J.; He, Q.; Wu, Y.; Zhang, C.; Li, X.; Yang, X.; Wu, Z.; Kang, J. Synthesis of Wafer-Scale Monolayer WS₂ Crystals toward the Application in Integrated Electronic Devices. *ACS Appl. Mater. Interfaces* **2019**, *11* (21), 19381–19387.

(13) Liu, X.; Hu, J.; Yue, C.; Della Fera, N.; Ling, Y.; Mao, Z.; Wei, J. High Performance Field-Effect Transistor Based on Multilayer Tungsten Disulfide. *ACS Nano* **2014**, *8* (10), 10396–10402.

(14) Fang, M.; Wang, F.; Han, Y.; Feng, Y.; Ren, T.; Li, Y.; Tang, D.; Song, Z.; Zhang, K. Controlled Growth of Bilayer-MoS₂ Films and MoS₂-Based Field-Effect Transistor (FET) Performance Optimization. *Adv. Electron. Mater.* **2018**, *4* (4), 1700524.

(15) Liu, L.; Li, T.; Ma, L.; Li, W.; Gao, S.; Sun, W.; Dong, R.; Zou, X.; Fan, D.; Shao, L.; Gu, C.; Dai, N.; Yu, Z.; Chen, X.; Tu, X.; Nie, Y.; Wang, P.; Wang, J.; Shi, Y.; Wang, X. Uniform Nucleation and Epitaxy of Bilayer Molybdenum Disulfide on Sapphire. *Nature* **2022**, *605* (7908), 69–75.

(16) Wang, J.; Liu, Y.; Chen, P.; Chuang, M.; Pezeshki, A.; Ling, D.; Chen, J.; Chen, Y.; Lee, Y. Controlled Low-Frequency Electrical Noise of Monolayer MoS₂ with Ohmic Contact and Tunable Carrier Concentration. *Adv. Electron. Mater.* **2018**, *4* (1), 1700340.

(17) Miakota, D. I.; Ghimire, G.; Kumar Ulaganathan, R.; Rodriguez, M. E.; Canulescu, S. A Novel Two-Step Route to Unidirectional Growth of Multilayer MoS₂ Nanoribbons. *Appl. Surf. Sci.* **2023**, *619* (October 2022), 156748.

(18) Li, X.; Li, B.; Lei, J.; Bets, K. V.; Sang, X.; Okogbue, E.; Liu, Y.; Unocic, R. R.; Jakobson, B. I.; Hone, J.; Harutyunyan, A. R. Nickel Particle-Enabled Width-Controlled Growth of Bilayer Molybdenum Disulfide Nanoribbons. *Sci. Adv.* **2021**, *7* (50), 1–9.

(19) Li, S.; Lin, Y.-C.; Zhao, W.; Wu, J.; Wang, Z.; Hu, Z.; Shen, Y.; Tang, D.-M.; Wang, J.; Zhang, Q.; Zhu, H.; Chu, L.; Zhao, W.; Liu, C.; Sun, Z.; Taniguchi, T.; Osada, M.; Chen, W.; Xu, Q.-H.; Wee, A. T. S.; Suenaga, K.; Ding, F.; Eda, G. Vapour-Liquid-Solid Growth of Monolayer MoS₂ Nanoribbons. *Nat. Mater.* **2018**, *17* (6), 535–542.

(20) Chen, Y. Growth of a Large, Single-Crystalline WS₂ Monolayer for High-Performance Photodetectors by Chemical Vapor Deposition. *Micromachines* **2021**, *12* (2), 137.

(21) Ren, D.-D.; Qin, J.-K.; Li, Y.; Miao, P.; Sun, Z.-Y.; Xu, P.; Zhen, L.; Xu, C.-Y. Photoluminescence Inhomogeneity and Excitons in CVD-Grown Monolayer WS₂. *Opt. Mater.* **2018**, *80* (May), 203–208.

(22) Bhattacharya, S.; Veluthandath, A. V.; Huang, C. C.; Murugan, G. S.; Bisht, P. B. Effect of Coating Few-Layer WS₂ on the Raman Spectra and Whispering Gallery Modes of a Microbottle Resonator. *J. Opt.* **2020**, *22* (10), 105003.

(23) Kaupmees, R.; Grossberg, M.; Ney, M.; Asaithambi, A.; Lorke, A.; Krustok, J. Tailoring of Bound Exciton Photoluminescence Emission in WS₂ Monolayers. *Phys. Status Solidi RRL* **2020**, *14* (2), 1900355.

(24) Huang, F.; Li, J. Z.; Xu, Z. H.; Liu, Y.; Luo, R. P.; Zhang, S. W.; Nie, P. B.; Lv, Y. F.; Zhao, S. X.; Su, W. T.; et al. A Bilayer 2D-WS₂/Organic-Based Heterojunction for High-Performance Photodetectors. *Nanomaterials* **2019**, *9* (9), 1312.

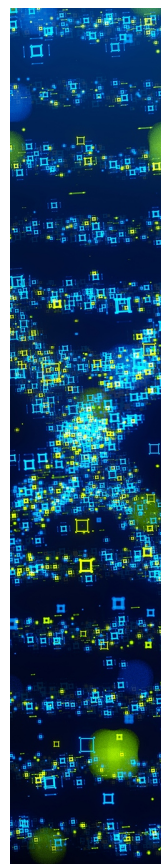
(25) Chen, F.; Ding, S.; Su, W. A Feasible Approach to Fabricate Two-Dimensional WS₂ Flakes: From Monolayer to Multilayer. *Ceram. Int.* **2018**, *44* (18), 22108–22112.

(26) He, Z.; Sheng, Y.; Rong, Y.; Lee, G.-D.; Li, J.; Warner, J. H. Layer-Dependent Modulation of Tungsten Disulfide Photoluminescence by Lateral Electric Fields. *ACS Nano* **2015**, *9* (3), 2740–2748.

(27) Li, S. Salt-Assisted Chemical Vapor Deposition of Two-Dimensional Transition Metal Dichalcogenides. *iScience* **2021**, *24* (11), 103229.

(28) Liu, D. S. H.; Hilse, M.; Lupini, A. R.; Redwing, J. M.; Engel-Herbert, R. Growth of Nanometer-Thick γ -InSe on Si(111) 7×7 by

Molecular Beam Epitaxy for Field-Effect Transistors and Optoelectronic Devices. *ACS Appl. Nano Mater.* **2023**, *6* (16), 15029–15037.



CAS BIOFINDER DISCOVERY PLATFORM™

**STOP DIGGING
THROUGH DATA
—START MAKING
DISCOVERIES**

CAS BioFinder helps you find the
right biological insights in seconds

Start your search

CAS
A Division of the
American Chemical Society